EPIB664 HW1

Alex Caramanico

2025-02-12

Load Libraries

```
library(ggplot2)
library(tidyverse)
library(stats)
library(cowplot)
library(survey)
```

Lecture 1 Problems

1

For Example 1.2, plot the scatter plot of X and Y including both observed and removed values together in each of the scenarios (MCAR, MAR, and MNAR) in Fig. 2. Use different symbols and colors for observed and removed values. Fit the regression of y on x for both observed and removed values and overlay the two fitted lines. The example used 1000 data points, to make it easier, you can use less data points such as 200.

```
set.seed(1549)
n <- 200
X <- rnorm(n,1,1) %>% sort() # sorted to make problem 2 easier
eps <- rnorm(n)
Y <- -2 + X + eps

# 1 minus, because P(X > c) = 1 - P(X < c)
mcar <- 1 - rbinom(n,1,0.4)
mar <- 1 - rbinom(n,1,exp(-1.4+X)/(1+exp(-1.4+X)))
mnar <- 1 - rbinom(n,1, exp(0.2+0.2*X+Y)/(1+exp(0.2+0.2*X+Y)))
all_data <- data.frame(X,Y,mcar,mar,mnar)

# MCAR
mcar_plot <- ggplot(all_data,mapping=aes(x=X,y=Y,color=factor(mcar))) +
  geom_point() +
  geom_smooth(method = "lm") +
  guides(color=guide_legend(title="Observed")) +
  ggtitle("MCAR - Missing Completely at Random")

# MAR
mar_plot <- ggplot(all_data,mapping=aes(x=X,y=Y,color=factor(mar))) +
```

```r
  geom_point() +
  geom_smooth(method = "lm") +
  guides(color=guide_legend(title="Observed")) +
  ggtitle("MAR - Missing at Random")

# MNAR
mnar_plot <- ggplot(all_data,mapping=aes(x=X,y=Y,color=factor(mnar))) +
  geom_point() +
  geom_smooth(method = "lm") +
  guides(color=guide_legend(title="Observed")) +
  ggtitle("MNAR - Missing Not at Random")

plot_grid(mcar_plot,mar_plot,mnar_plot)
```


```r
# All red points are the "missing" observations. The plots reveal that the MCAR
# and MAR mechanisms have only slight effects on the relationship between the
# variables, whereas MNAR alters the regression line considerably. To be
# specific, the regression line for missing data from the MNAR mechanism has a
# higher intercept, because higher values of X or Y increase the chance of the
# data being "missing". The smaller effect of the other two mechanisms is due to
# complete randomness (hence the name) for MCAR, and because only ONE of the
# variables has an effect on missingness for MAR.

# It should also be noted that these results can vary because of the use of
# random number generation or if different sample sizes are used.
```

```r
# Testing if MCAR or not MCAR
data.frame(MCAR=c(pval=t.test(subset(all_data,all_data$mcar==1)$X,
                              subset(all_data,all_data$mcar==0)$X)$p.value,
                  t.test(subset(all_data,all_data$mcar==1)$X,
                         subset(all_data,all_data$mcar==0)$X)$estimate),
           MAR=c(pval=t.test(subset(all_data,all_data$mar==1)$X,
                             subset(all_data,all_data$mar==0)$X)$p.value,
                 t.test(subset(all_data,all_data$mar==1)$X,
                        subset(all_data,all_data$mar==0)$X)$estimate),
           MNAR=c(pval=t.test(subset(all_data,all_data$mnar==1)$X,
                              subset(all_data,all_data$mnar==0)$X)$p.value,
                  t.test(subset(all_data,all_data$mnar==1)$X,
                         subset(all_data,all_data$mnar==0)$X)$estimate))
```

```
##                MCAR          MAR         MNAR
## pval      0.8134936 8.732129e-09 2.357475e-08
## mean of x 1.0336481 6.528233e-01 6.575126e-01
## mean of y 0.9939177 1.515695e+00 1.509351e+00
```

```r
# The first one is of course likely MCAR by the t test, while the others likely
# are NOT MCAR due to the very small p-values.

rm(eps,mcar,mnar,all_data,mcar_plot,mnar_plot,mar_plot)
```

2

In Example 1.2, the response function Pr(R = 1|X) is a logistic regression of X. Can you generate MAR and MNAR not using the logistic regression? Then test if the generated missing data are MCAR in both scenarios and report the results. Hint: you can do it by generate missing data using different probability at different ranges of X and Y. For example, you can divide subjects into two groups using the median of X; for any subject whose X-value is less than the median, the probability of missingness of Y is 30%; and for any subject whose X-value is larger than the median, the probability of missingness is 50%. Overall it would have Y missing around 40%. But still as X increases, the missingness of Y increases. This would create a MAR scenario.

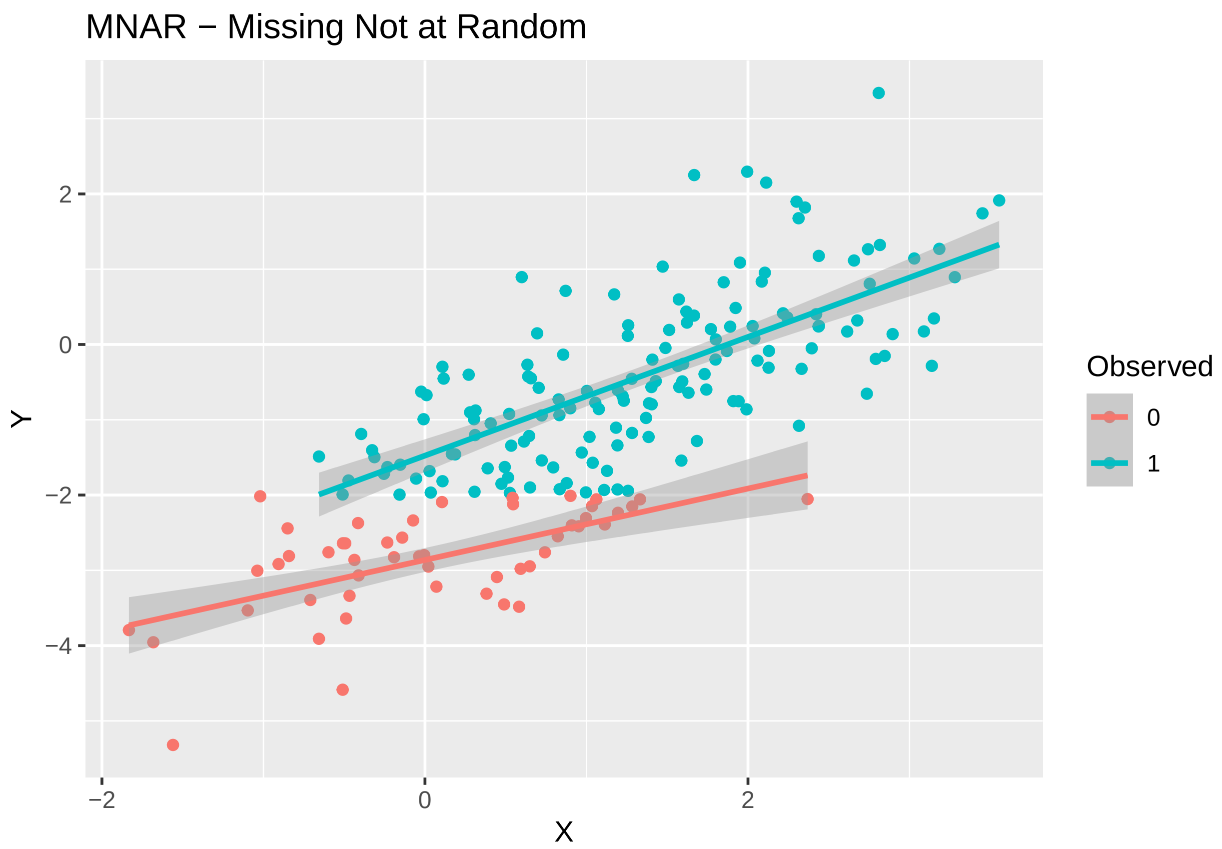
```r
# MAR - missingness depends on values of x ONLY
# *using sorted data frame from Problem 1*
mar <- c(1-rbinom(n/2,1,0.3), # first half missing with 30% chance
         1-rbinom(n/2,1,0.5)) # second half missing with 50% chance

# verification
# cbind(X,Y,mar)[1:100,] %>% colMeans() # about 30% missing
# cbind(X,Y,mar)[101:200,] %>% colMeans() # about 50%  missing

all_data2 <- data.frame(X,Y,mar)

observed_X <- subset(all_data2,all_data2$mar==1)$X
missing_X <- subset(all_data2,all_data2$mar==0)$X

c(pval=t.test(observed_X,missing_X)$p.value,
  t.test(observed_X,missing_X)$estimate)
```

```
##        pval   mean of x   mean of y
## 0.003356103 0.841477915 1.310072434
```

```
# Means are significantly different, MCAR unlikely. This result can vary a lot
# with different runs of the same experiment, but it is usually significant

ggplot(all_data2,mapping=aes(x=X,y=Y,color=factor(mar))) +
  geom_point() +
  geom_smooth(method = "lm") +
  guides(color=guide_legend(title="Observed")) +
  ggtitle("MAR - Missing at Random")
```


```
# slightly higher proportion of missing values when X increases

# MNAR - missingness depends on values of x AND y
# we'll say Y's less than -2 are all missing
# X and Y are positively correlated by their definition in the set-up, so taking
# away the tail of either variable will greatly affect the mean of the other
all_data2 <- cbind(all_data2,
                   mnar = ifelse(all_data2$Y > -2,
                                 1,0))

ggplot(all_data2,mapping=aes(x=X,y=Y,color=factor(mnar))) +
  geom_point() +
  geom_smooth(method = "lm") +
```

```
  guides(color=guide_legend(title="Observed")) +
  ggtitle("MNAR - Missing Not at Random")
```




```
observed_X <- subset(all_data2,all_data2$mnar==1)$X
missing_X <- subset(all_data2,all_data2$mnar==0)$X
c(pval=t.test(observed_X,missing_X)$p.value,
  t.test(observed_X,missing_X)$estimate)
```

```
##         pval    mean of x    mean of y
## 6.620186e-14 1.346467e+00 3.877548e-02
```

```
rm(n,all_data2,observed_X,missing_X)
```

3

Identify a real dataset and a univariate missing data problem with one missing variable (Y or some other names you prefer) and at least two fully-observed variables (X1, X2, etc or some other names you prefer). Report the missingness rate of the incomplete variable. Which mechanism might it be: MCAR, MAR, or MNAR? Use some analytical results to support your arguments.

```
data("airquality")
colSums(is.na(airquality)) / nrow(airquality)
```

```
##      Ozone    Solar.R       Wind       Temp      Month        Day
## 0.24183007 0.04575163 0.00000000 0.00000000 0.00000000 0.00000000
```

```r
# Two fully observed variables: Wind, Temp
# 24.18% of "Ozone" missing, 4.5% of "Solar.R" missing

# My initial guess, before any analysis, would be MCAR or possibly MAR. I would
# MCAR instead of MAR because the NAs for Ozone are more likely due to the
# measurement just not being taken, or being unavailable, regardless of what the
# wind or temperature was. MNAR can be ruled out for a similar reason - I do not
# anticipate Ozone being missing due to it taking on a particular value (i.e.
# very small or very large). However, it could be possible that the tools used
# to measure the variables are limited in the values they display, in which
# case MNAR is a possibility. It could also be that either of the complete
# variables have a strong correlation with Ozone, affecting its missingness.
# The analysis below attempts to better contextualize the problem.

reduced <- airquality[!is.na(airquality$Ozone),]
reduced <- reduced[!is.na(reduced$Solar.R),]
cor(reduced[,1:4])
```

```
##              Ozone    Solar.R       Wind       Temp
## Ozone    1.0000000  0.3483417 -0.6124966  0.6985414
## Solar.R  0.3483417  1.0000000 -0.1271835  0.2940876
## Wind    -0.6124966 -0.1271835  1.0000000 -0.4971897
## Temp     0.6985414  0.2940876 -0.4971897  1.0000000
```

```r
# Ozone has a possibly considerable relationship with both Wind and Temp, with a
# moderate negative and moderate positive correlation with each respectively. I
# would argue that this is a logical necessity for MAR to be the missingness
# mechanism at play.

missing_o <- subset(airquality,airquality$Ozone %>% is.na())
observed_o <- subset(airquality,!airquality$Ozone %>% is.na())
c(pval=t.test(missing_o$Wind,observed_o$Wind)$p.value,
  t.test(missing_o$Wind,observed_o$Wind)$estimate)
```

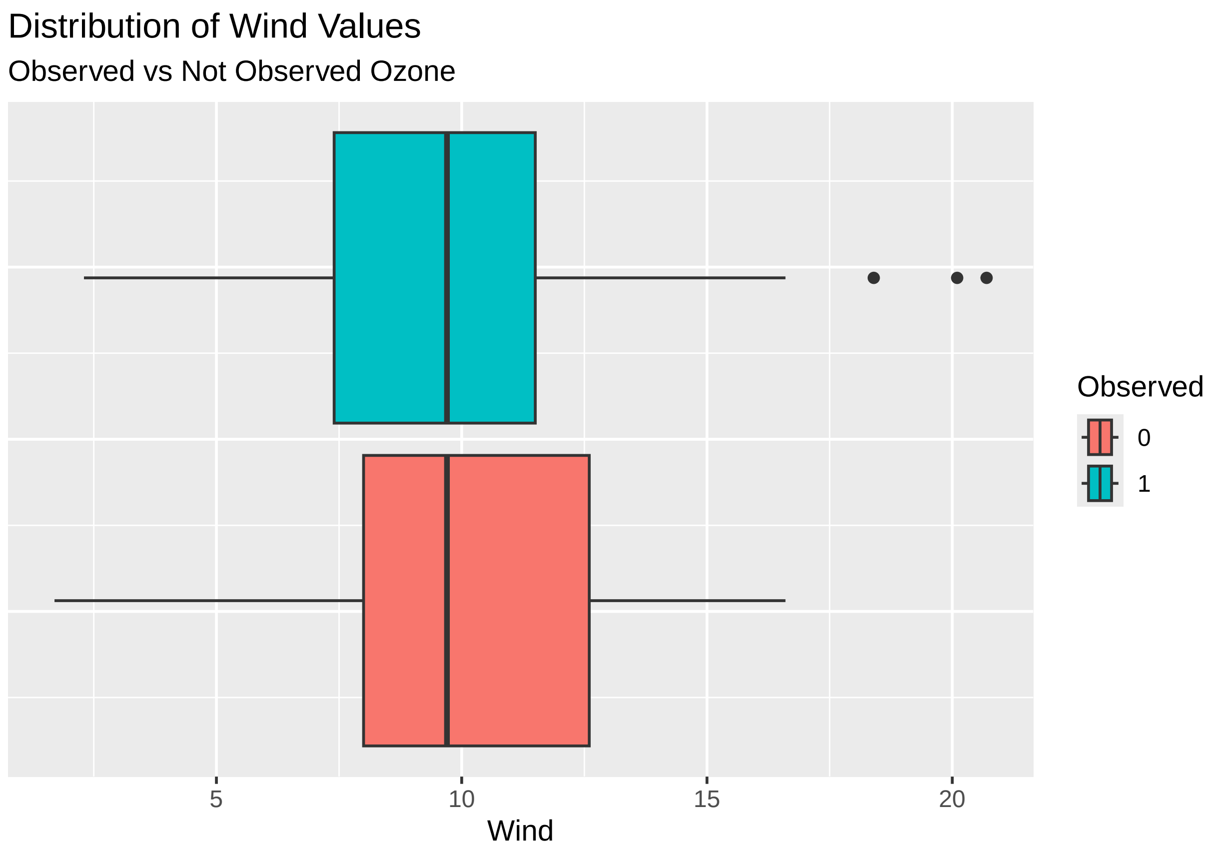
```
##       pval  mean of x  mean of y
##  0.5446224 10.2567568  9.8620690
```

```r
c(pval=t.test(missing_o$Temp,observed_o$Temp)$p.value,
  t.test(missing_o$Temp,observed_o$Temp)$estimate)
```

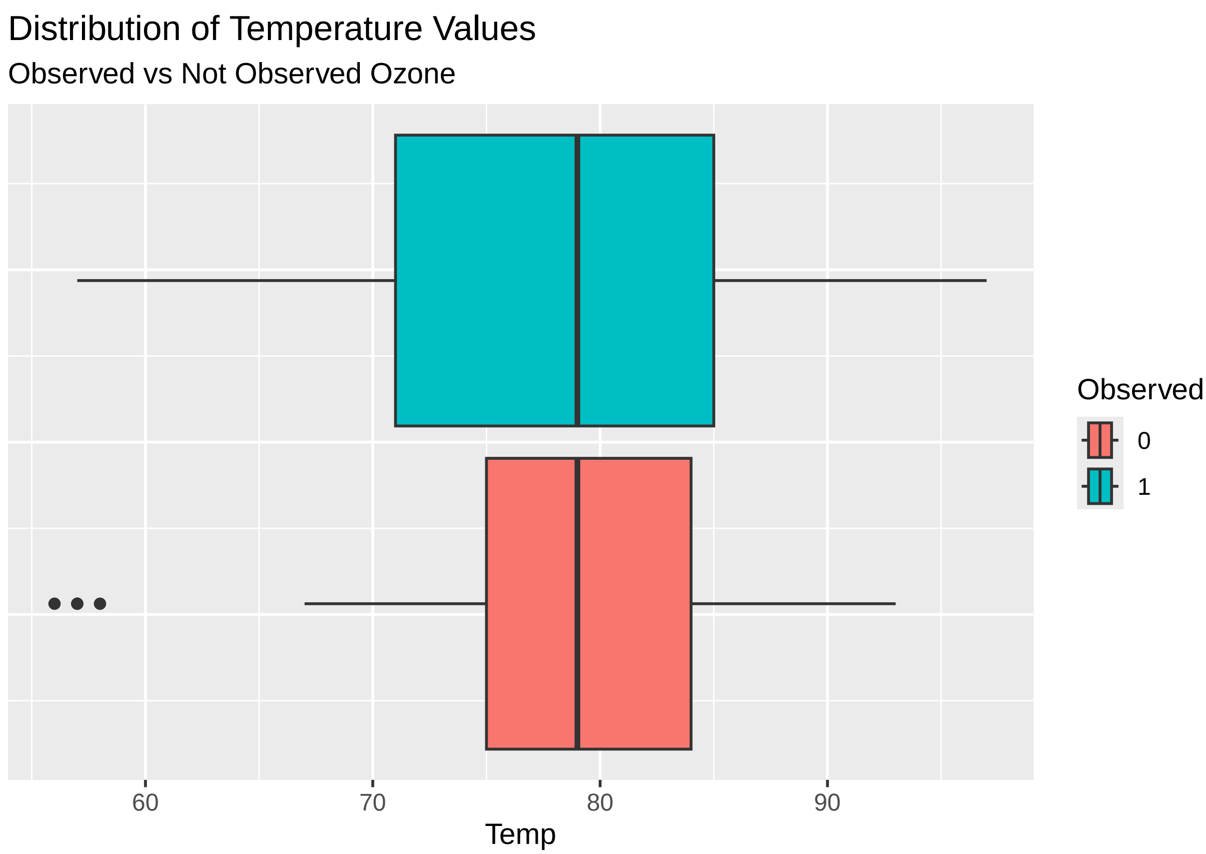
```
##       pval  mean of x  mean of y
##  0.9786832 77.9189189 77.8706897
```

```r
# Means of either fully observed variable not significantly different between
# missing and observed ozone values suggests that MCAR is likely. These results
# seem to imply that the Ozone variable has missing values completely at random,
# or in other words, its missingness does not depend on the values of any
# observed variables or its own value.
```

```r
airquality$Observed <- ifelse(is.na(airquality$Ozone),0,1) %>% as.factor()
ggplot(data=airquality) +
  geom_boxplot(mapping=aes(x=Wind,fill=Observed)) +
  theme(axis.text.y=element_blank(),axis.ticks.y=element_blank()) + # remove Y axis
  ggtitle("Distribution of Wind Values", "Observed vs Not Observed Ozone")
```




```r
# The distribution of the wind variable seems to be pretty similar regardless
# of whether or not the corresponding Ozone value is missing, aside from slight
# differences in upper/lower quartiles and the observed values having a couple
# outliers.

ggplot(data=airquality) +
  geom_boxplot(mapping=aes(x=Temp,fill=Observed)) +
  theme(axis.text.y=element_blank(),axis.ticks.y=element_blank()) + # remove Y axis
  ggtitle("Distribution of Temperature Values", "Observed vs Not Observed Ozone")
```




```r
# The difference in temperature values between the observed and non-observed
# ozone groups is more apparent than for the wind variable, although there is
# not any extreme difference. The range is very comparable, but the unobserved
# group would have a tighter (less varied) distribution around its center because
# the upper and lower quartiles are closer to the mean. However, the mean value
# is very similar between the groups, corroborating the result from the t test.
# I would argue that there is not enough evidence in either of these plots to
# suggest the mechanism is anything other than MCAR. If there is some relation
# between either of these variables and the missingness of Ozone, it is likely
# extremely subtle and difficult to detect (at least with the data available).

rm(reduced,missing_o,observed_o)
```

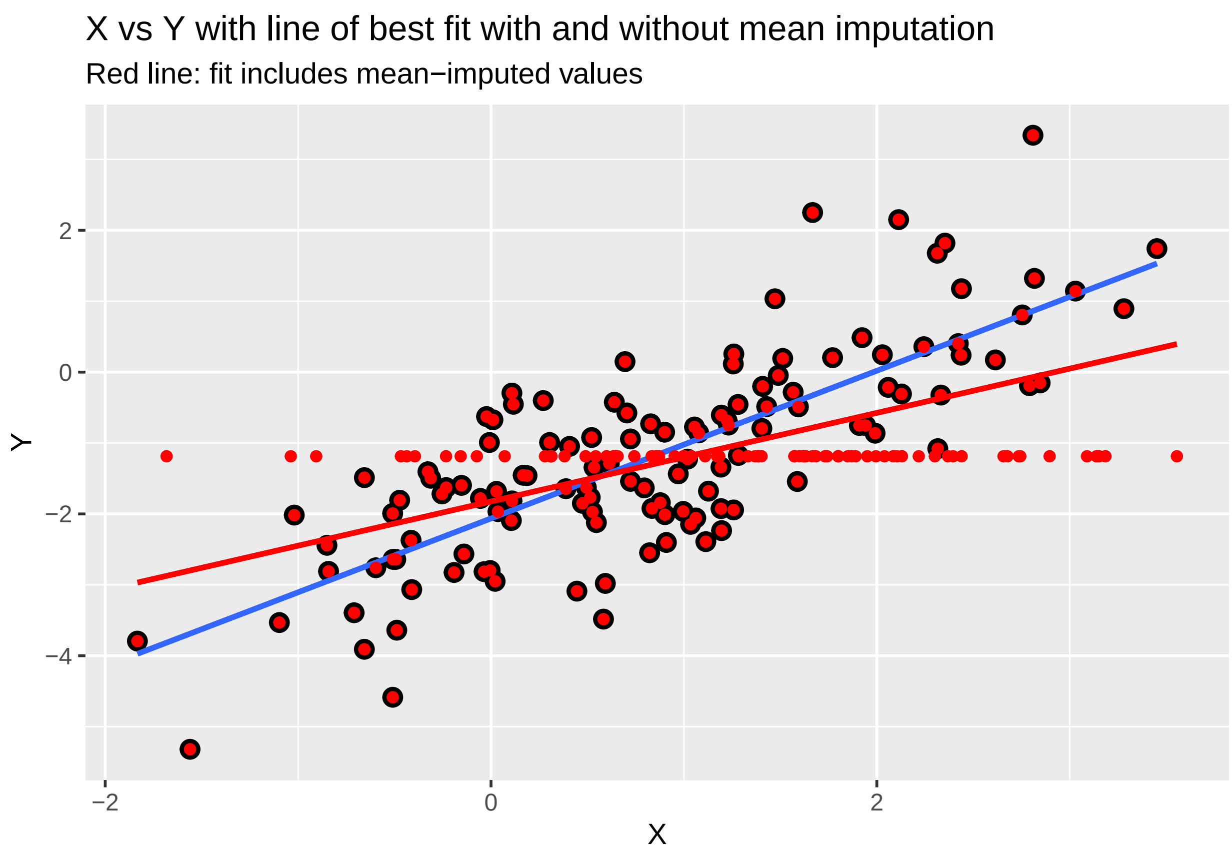
Lecture 2 Problems

1

Use simulated data (Example 1.2) to show why using the unconditional mean imputation can lead to biased mean and regression estimate if the missing variable is MAR; and the bias of the regression is towards the null value (attenuated).

```r
Y_mar <- ifelse(mar,Y,NA) # if missing, i.e. mar = 0, change Y to NA
Y_mar_impute <- ifelse(mar,Y_mar,
                       mean(Y_mar,na.rm=TRUE)) # impute mean for missing values
```

```r
imputation <- data.frame(X,Y,mar,Y_mar,Y_mar_impute)
ggplot(data=imputation) +
  geom_point(mapping=aes(x=X,y=Y_mar),size=3) +
  geom_smooth(mapping=aes(x=X,y=Y_mar),method='lm',se=F) +
  geom_point(mapping=aes(x=X,y=Y_mar_impute),color='red') +
  geom_smooth(mapping=aes(x=X,y=Y_mar_impute),method='lm',se=F,color='red') +
  ylab("Y") +
  ggtitle("X vs Y with line of best fit with and without mean imputation",
          "Red line: fit includes mean-imputed values")
```




```r
summary(lm(Y_mar~X))$coefficients
```

```
##               Estimate Std. Error   t value     Pr(>|t|)
## (Intercept) -2.063634 0.10281344 -20.07163 1.815639e-40
## X            1.041536 0.07497284  13.89217 6.818763e-27
```

```r
summary(lm(Y_mar~X))$r.squared
```

```
## [1] 0.6126891
```

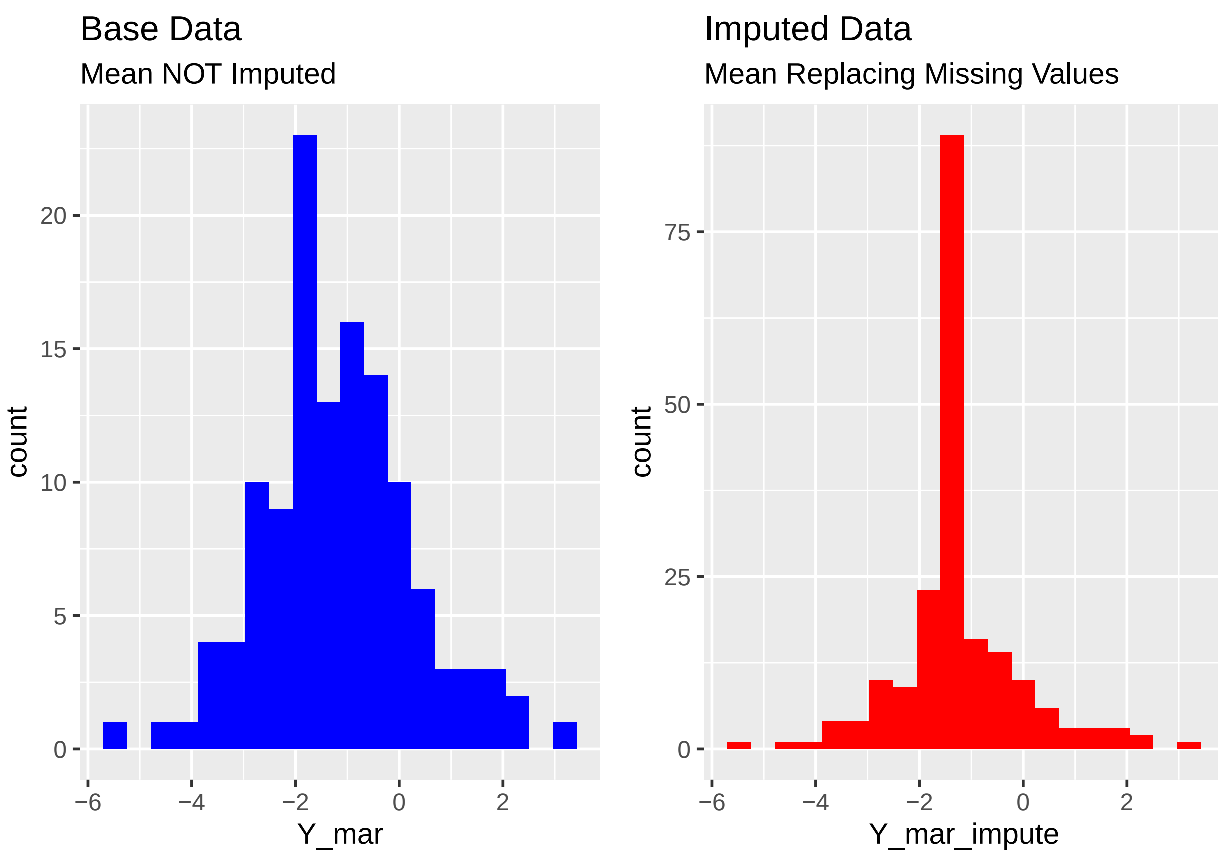
```r
summary(lm(Y_mar_impute~X))$coefficients
```

```
##                Estimate Std. Error   t value     Pr(>|t|)
## (Intercept) -1.8236998 0.08739905 -20.86636 6.868607e-52
## X            0.6242942 0.05823673  10.71994 1.946574e-21
```

```r
summary(lm(Y_mar_impute~X))$r.squared
```

```
## [1] 0.3672445
```

```r
# The summary tables and lines of best fit demonstrate that mean imputation
# biases the regression estimates, specifically by bringing the slope estimate
# closer to 0. This can be problematic in real-world examples, as data with a
# large number of missing values could be determined to have no relationship
# between its variables if this method is utilized. Though the significance of
# the regression holds in this example, it may not in others, and in any case
# the estimates undershoot the true effect.
# Additionally, the R^2 value is significantly worse in the imputation example,
# which not only corroborates the previous arguments (in that the independent
# variable has less explanatory power) but dramatically reduces any potential
# model utility.
# The results make logical sense as well just from looking at the scatter plot.
# The straight line of red dots in the plot - across many X values - illustrates
# the point. If we assign the same Y value across various X values (which follow
# the missing at random mechanism), we're ignoring any potential relationship
# that may exist between the variables. Y is missing at random, so it is not
# as if only small or only large values of Y are missing. Imputing the mean
# pushes everything toward the center, violating any assumption of a relationship
# between the variables and mathematically reducing said relationship.

miss <- ggplot(data=imputation) +
  geom_histogram(mapping=aes(x=Y_mar),fill='blue',bins=20) +
  ggtitle("Base Data","Mean NOT Imputed")

imp <- ggplot(data=imputation) +
  geom_histogram(mapping=aes(x=Y_mar_impute),fill="red",bins=20) +
  ggtitle("Imputed Data", "Mean Replacing Missing Values")
plot_grid(miss,imp)
```




```r
c(Base_Var=var(imputation$Y_mar,na.rm=T),
  Impute_Var=var(imputation$Y_mar_impute))
```

```
##   Base_Var Impute_Var
##   2.092838   1.293563
```

```r
# This plot demonstrates another issue with mean imputation - it artificially
# reduces the variance of the missing variable dramatically. Though the means
# of the two sets of response variables are the same (of course), the variance
# is much lower for the imputed set because it naturally has *many* repeated/
# identical values. This is problematic because the new variance likely is not
# reflective of the true variance of the population from which the sample was
# taken.

rm(imputation,mar,n,X,Y,Y_mar,Y_mar_impute,miss,imp)
```

2

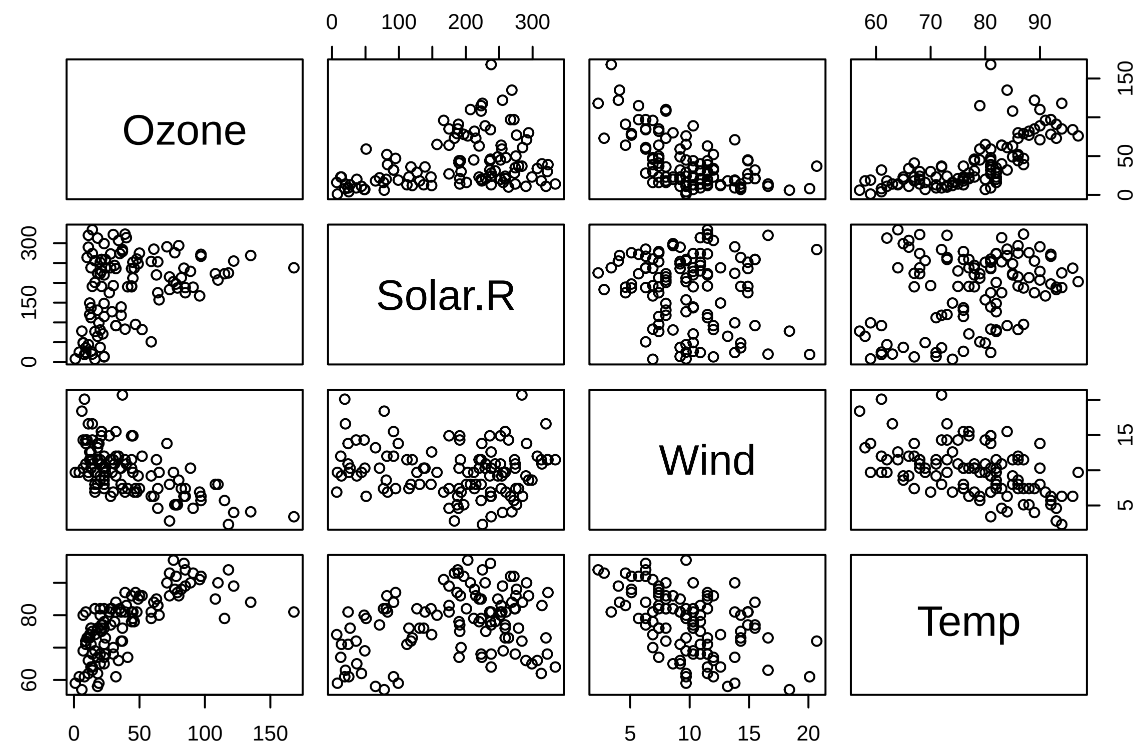
Identify a real dataset and a univariate missing data problem with one missing variable (y or some names you prefer) and at least two fully-observed variables (x1, x2, etc or some names you prefer). Try some of the discussed ad-hoc missing data methods (e.g.,complete-case analysis, mean imputation, regression prediction, missingness indicator method when appropriate) in the class and compare their estimates (e.g., for the mean of the incomplete variable or some regression relationship between the incomplete and observed variables). Report the results and add some discussion.

Complete-Case Analysis

```r
# I will use the airquality data set again, focusing primarily on the Ozone
# variable and its relationship with the temperature and wind covariates. Since
# Ozone is a quantitative variable rather than a categorical one, the missingness
# indicator method likely would not be very meaningful.
data("airquality")
base_results <- summary(lm(airquality$Ozone~
                             airquality$Wind+airquality$Temp)) # base data

cca_data <- na.omit(airquality)
cca_results <- summary(lm(cca_data$Ozone~cca_data$Wind+cca_data$Temp))
cca_results$coefficients
```

```
##                 Estimate Std. Error   t value     Pr(>|t|)
## (Intercept)   -67.321953 23.6210451 -2.850084 5.236066e-03
## cca_data$Wind  -3.294839  0.6711482 -4.909257 3.261728e-06
## cca_data$Temp   1.827554  0.2505520  7.294112 5.294630e-11
```

```r
plot(cca_data[,1:4])
```

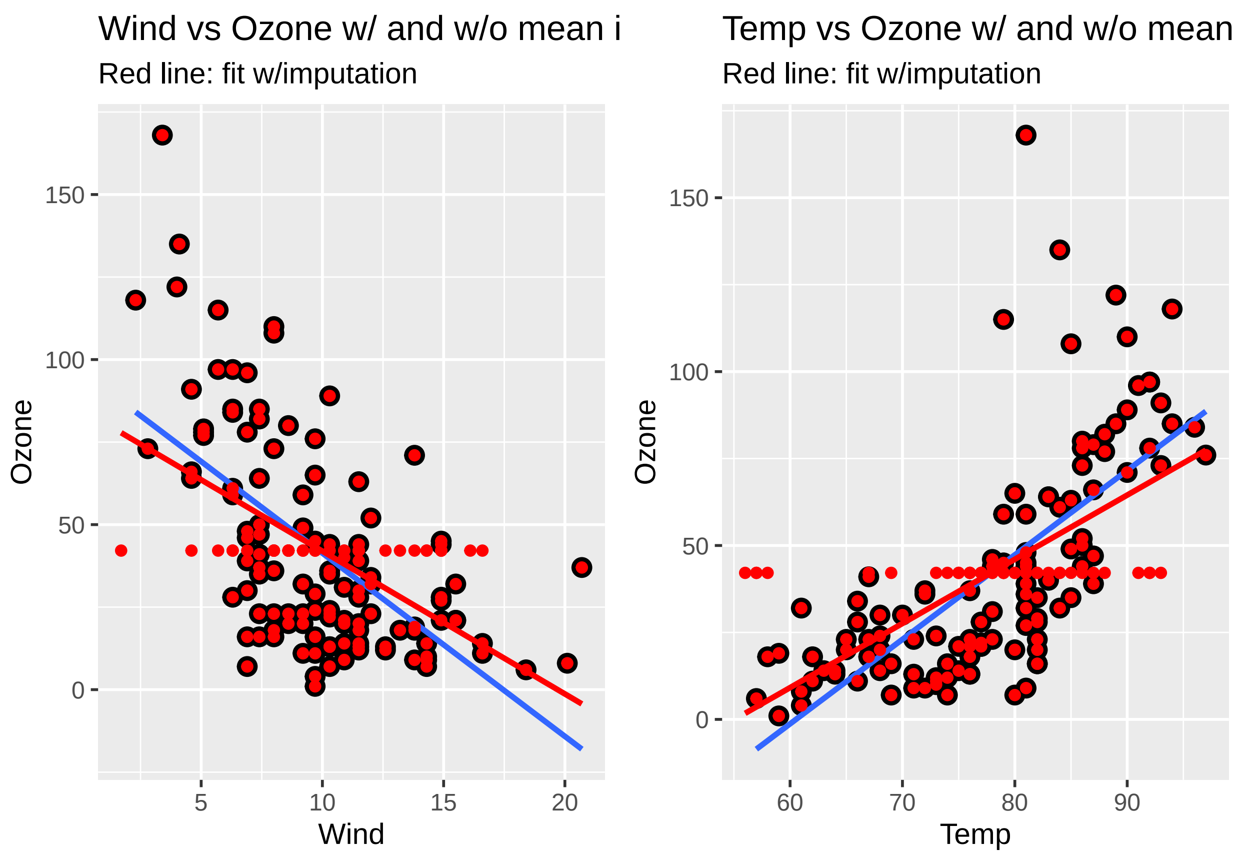



```r
# The complete-case approach reduces the sample size from 153 to 111, a 27.4%
# reduction. Note that this removed any rows where the Solar.R variable is
# missing as well as those with missing Ozone values. I fit a model with only
```

```r
# Temperature and Wind as the explanatory variables, even though Solar.R had no
# missing values. This will allow for direct comparisons to the other approaches.
# The regression shows a significant relationship between Ozone and
# both of the explanatory variables, with temperature having the strongest
# according to the test statistics and p-values. Notably, wind has a negative
# relationship with ozone. That is, the ozone value decreases as wind increases,
# whereas ozone increases when Solar.R or temperature increase, but much more
# strongly for the latter. All of this is reflected by the plot as well, which
# demonstrates slight relationships for Solar.R and Wind but a strong relation
# for temperature, all compared to Ozone.
# Another point to consider when moving forward in this analysis is collinearity
# between the explanatory variables. From the scatter plots, Solar.R and Wind
# and Solar.R and Temperature do not appear to have much of a relationship.
# However, Wind and Temperature seem to have a fairly considerable negative
# correlation, which could have adverse effects on the regression estimates (but
# is not the point of this exercise).
rm(cca_data)
```

Mean Imputation

```r
ozone_impute <- ifelse(!is.na(airquality$Ozone),airquality$Ozone,
                       mean(airquality$Ozone,na.rm=T))
impute_data <- cbind(airquality,ozone_impute)
# the following plots look at Wind and Temperature independently
wind_smooth <- ggplot(data=impute_data) +
  geom_point(mapping=aes(x=Wind,y=Ozone),size=3) +
  geom_smooth(mapping=aes(x=Wind,y=Ozone),method='lm',se=F) +
  geom_point(mapping=aes(x=Wind,y=ozone_impute),color='red') +
  geom_smooth(mapping=aes(x=Wind,y=ozone_impute),method='lm',se=F,color='red') +
  ggtitle("Wind vs Ozone w/ and w/o mean imputation",
          "Red line: fit w/imputation")

temp_smooth <- ggplot(data=impute_data) +
  geom_point(mapping=aes(x=Temp,y=Ozone),size=3) +
  geom_smooth(mapping=aes(x=Temp,y=Ozone),method='lm',se=F) +
  geom_point(mapping=aes(x=Temp,y=ozone_impute),color='red') +
  geom_smooth(mapping=aes(x=Temp,y=ozone_impute),method='lm',se=F,color='red') +
  ggtitle("Temp vs Ozone w/ and w/o mean imputation",
          "Red line: fit w/imputation")

plot_grid(wind_smooth,temp_smooth)
```




```
c(Base_Var=var(impute_data$Ozone,na.rm=T),Impute_Var=var(impute_data$ozone_impute))
```

```
##   Base_Var Impute_Var
##  1088.2005   823.3096
```

```
# For both plots we see a similar result to the previous problem - the imputed
# data leads to an attenuated regression estimate (closer to null relationship
# between the variables). The variance of the Ozone variable is greatly reduced
# due to mean imputation, also similar to before.

impute_results <- summary(lm(impute_data$ozone_impute~
                               impute_data$Wind+impute_data$Temp)) # imputed
base_results$coefficients
```

```
##                    Estimate Std. Error   t value     Pr(>|t|)
## (Intercept)      -71.033218 23.5779922 -3.012692 3.196240e-03
## airquality$Wind   -3.055491  0.6632503 -4.606844 1.080046e-05
## airquality$Temp    1.840179  0.2499634  7.361793 3.149109e-11
```

```
base_results$r.squared
```

```
## [1] 0.5687097
```

```r
impute_results$coefficients
```

```
##                     Estimate Std. Error   t value     Pr(>|t|)
## (Intercept)       -41.215871 19.3089752 -2.134545 3.442334e-02
## impute_data$Wind   -2.598643  0.5542705 -4.688401 6.137230e-06
## impute_data$Temp    1.402387  0.2063011  6.797768 2.350451e-10
```

```r
impute_results$r.squared
```

```
## [1] 0.4510154
```

```r
# The results of the regressions are similar regardless of mean imputation in
# that both of the independent variables are significant in explaining the
# variance in Ozone. However, like the last problem, the R2 value is much lower
# in the imputed data case because of the attenuation. The relationship between
# wind and ozone AND temperature and ozone is negatively biased, in other words
# the parameter estimates for the slopes are beneath the expected true values.

rm(ozone_impute,impute_data,wind_smooth,temp_smooth)
```

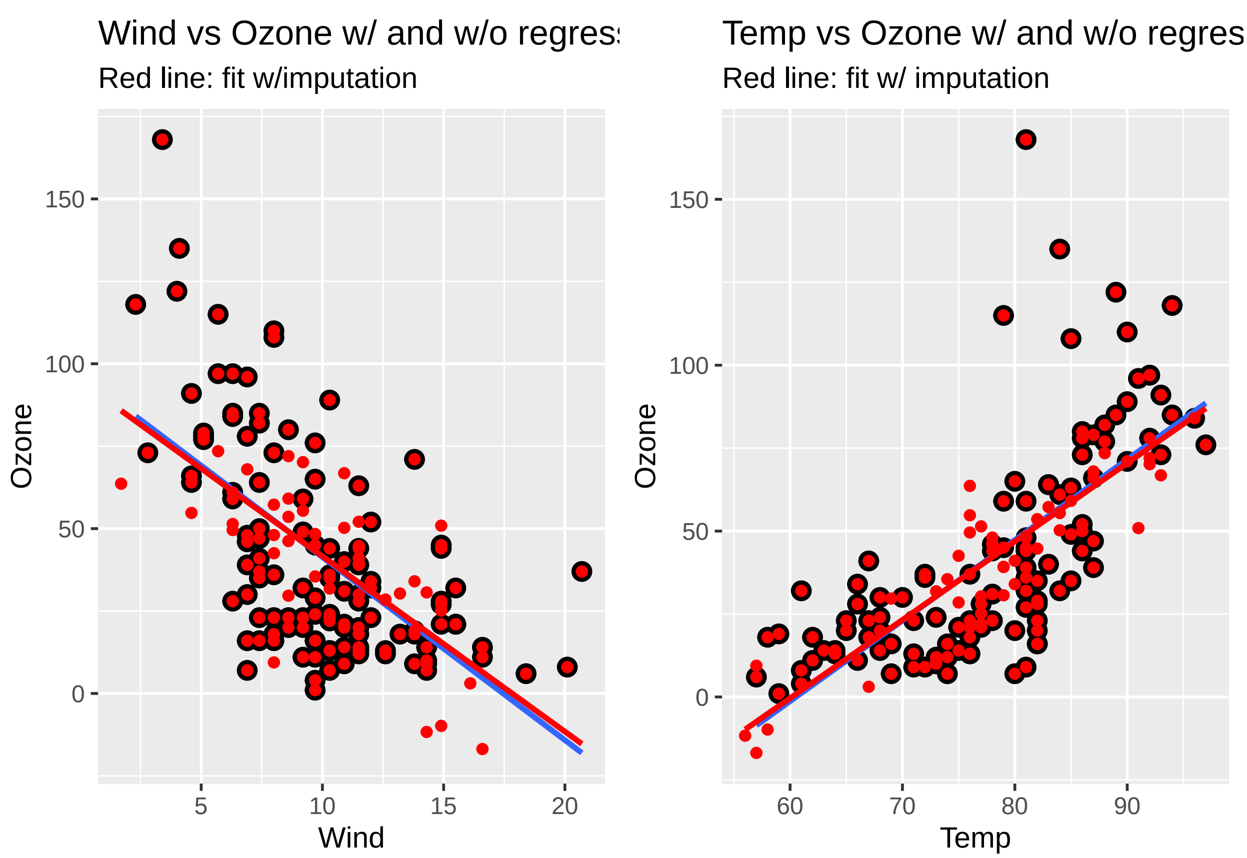
Regression Prediction

```r
intercept <- base_results$coefficients[,1][1]
slopes <- base_results$coefficients[,1][2:3]
fitted <- airquality$Wind * slopes[1] + airquality$Temp * slopes[2] + intercept

fitted_impute_data <- cbind(airquality,
                            fit = ifelse(airquality$Ozone %>% is.na(),
                                         fitted,
                                         airquality$Ozone))
head(fitted_impute_data[,c(1,3,4,7)],10)
```

```
##    Ozone Wind Temp       fit
## 1     41  7.4   67  41.00000
## 2     36  8.0   72  36.00000
## 3     12 12.6   74  12.00000
## 4     18 11.5   62  18.00000
## 5     NA 14.3   56 -11.67673
## 6     28 14.9   66  28.00000
## 7     23  8.6   65  23.00000
## 8     19 13.8   59  19.00000
## 9      8 20.1   61   8.00000
## 10    NA  8.6   69  29.66190
```

```r
# The code above imputes the fitted value only if the Ozone value is missing. It
# should be noted that there are some negative fitted values, which does not
# make sense for the variable we're examining. This is because high winds and
# low temperatures mean the negative parameter estimate for the former outweighs
# the latter, pushing the estimated Ozone value below 0. There are only three
```

```r
# instances where this happened, and it could be desirable to set the ozone
# values to 0 but that is another form of data manipulation which could hurt the
# reliability of results even further.

fitted_impute_data$Observed <- ifelse(is.na(airquality$Ozone),0,1) %>% as.factor()
wind_smooth <- ggplot(data=fitted_impute_data) +
  geom_point(mapping=aes(x=Wind,y=Ozone),size=3) +
  geom_smooth(mapping=aes(x=Wind,y=Ozone),method='lm',se=F) +
  geom_point(mapping=aes(x=Wind,y=fit),color='red') +
  geom_smooth(mapping=aes(x=Wind,y=fit),method='lm',se=F,color='red') +
  ggtitle("Wind vs Ozone w/ and w/o regression imputation",
          "Red line: fit w/imputation")

temp_smooth <- ggplot(data=fitted_impute_data) +
  geom_point(mapping=aes(x=Temp,y=Ozone),size=3) +
  geom_smooth(mapping=aes(x=Temp,y=Ozone),method='lm',se=F) +
  geom_point(mapping=aes(x=Temp,y=fit),color='red') +
  geom_smooth(mapping=aes(x=Temp,y=fit),method='lm',se=F,color='red') +
  ggtitle("Temp vs Ozone w/ and w/o regression imputation",
          "Red line: fit w/ imputation")

plot_grid(wind_smooth,temp_smooth)
```




```r
# In a two-variable problem, we'd expect to see the red dots without black
# circles to follow a straight line, as they were calculated by the same linear
```

```r
# function. However since this is a multivariate problem, said line would be in
# three dimensions (there are three beta parameters). In any case, the regression
# line is extremely similar regardless of whether regression-imputed values are
# utilized, which makes sense because this technique simply adds points to where
# they'd be expected to go in the plots (for lack of better terminology).

regression_results <- summary(lm(fitted_impute_data$fit~
                                fitted_impute_data$Wind+fitted_impute_data$Temp))
# The most obvious issue with this approach is that it creates several points
# with 0 residual, which can be "ideal" in theory, but not when the values were
# not observed. This process does not account for the random noise affecting
# real observed values. Though this can easily be accounted for with additional
# random number generation, there is also the concern of fitted values falling
# outside the possible range (as seen in this example).

rm(fitted_impute_data,intercept,
   slopes,fitted,temp_smooth,wind_smooth)
```

Comparing All Estimates

```r
base_results$coefficients
```

```
##                    Estimate Std. Error   t value     Pr(>|t|)
## (Intercept)     -71.033218 23.5779922 -3.012692 3.196240e-03
## airquality$Wind  -3.055491  0.6632503 -4.606844 1.080046e-05
## airquality$Temp   1.840179  0.2499634  7.361793 3.149109e-11
```

```r
cca_results$coefficients
```

```
##                  Estimate Std. Error   t value     Pr(>|t|)
## (Intercept)   -67.321953 23.6210451 -2.850084 5.236066e-03
## cca_data$Wind  -3.294839  0.6711482 -4.909257 3.261728e-06
## cca_data$Temp   1.827554  0.2505520  7.294112 5.294630e-11
```

```r
impute_results$coefficients
```

```
##                     Estimate Std. Error   t value     Pr(>|t|)
## (Intercept)      -41.215871 19.3089752 -2.134545 3.442334e-02
## impute_data$Wind  -2.598643  0.5542705 -4.688401 6.137230e-06
## impute_data$Temp   1.402387  0.2063011  6.797768 2.350451e-10
```

```r
regression_results$coefficients
```

```
##                             Estimate Std. Error   t value     Pr(>|t|)
## (Intercept)               -71.033218 17.1144923 -4.150472 5.541491e-05
## fitted_impute_data$Wind    -3.055491  0.4912771 -6.219485 4.719372e-09
## fitted_impute_data$Temp     1.840179  0.1828548 10.063606 1.591674e-18
```

```r
# The parameter estimates for the base and regression-imputed data are of course
# identical because the latter simply added more points that fit exactly on the
# line described by said numbers. As a result, the standard errors and thus
# p-values are better because there are more points fitting the regression line
# well. Neither the significance nor the direction of the relationship changes
# for any variable across any of the methods - but perhaps the most notable
# thing here is again the fact that mean imputation has negatively biased
# estimates because of its reduction of Ozone's variance.
# *** The only reason complete-case analysis and the base results are different
# is that complete-case also removed any rows with missing Solar.R values, and
# thus fitted a model based on fewer Ozone observations. They would be exactly
# identical otherwise. ***

cbind(Base=base_results$adj.r.squared,
      Complete=cca_results$adj.r.squared,
      Mean=impute_results$adj.r.squared,
      Regression=regression_results$adj.r.squared)
```

```
##           Base  Complete      Mean Regression
## [1,] 0.5610762 0.5736257 0.4436956  0.6196724
```

```r
# It is clear that regression-imputation offers a considerable improvement in
# model fit as it relates to the R^2 value. Again, this is because more points
# are being added that fit the line well (exactly, for that matter). In a way
# this could almost be described as a "self-affirming" process, where the the
# model essentially makes itself better via this methodology. Mean imputation
# is of course the worst due to its attenuation of variable relationships
```

Lecture 3 Problems

1

Identify a real dataset and a univariate missing data problem with one missing variable (y) and at least two fully-observed variables (x_1, x_2, etc). Try the nonresponse weighting missing methods in the class and compare their estimates (e.g., for the mean of the incomplete variable or some regression relationship between the incomplete and observed variables) and also compare them with the ad-hoc missing data methods learned in Lecture 2.

If none of X-variables is related to the response (e.g., likely MCAR), then use some X variables you prefer to fit the propensity score model and obtain the response weights.

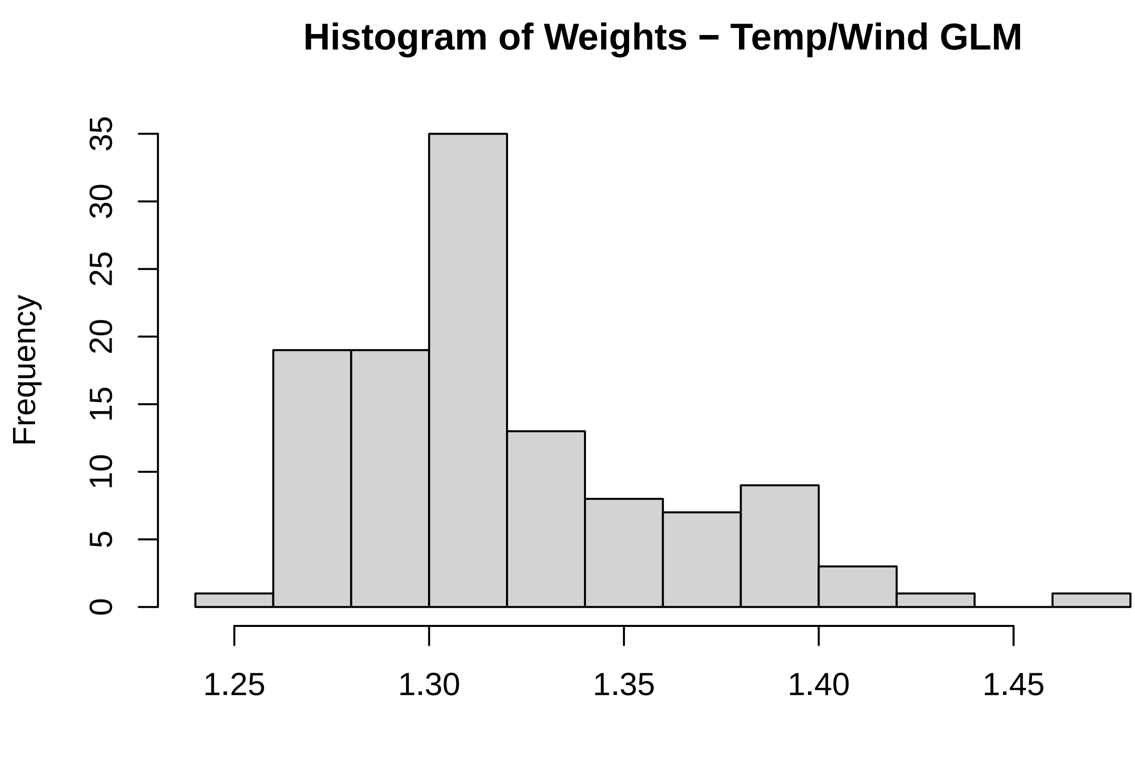
```r
data("airquality")
airquality$Observed <- ifelse(is.na(airquality$Ozone),0,1)
cor(na.omit(airquality)[1:4])
```

```
##              Ozone    Solar.R       Wind       Temp
## Ozone    1.0000000  0.3483417 -0.6124966  0.6985414
## Solar.R  0.3483417  1.0000000 -0.1271835  0.2940876
## Wind    -0.6124966 -0.1271835  1.0000000 -0.4971897
## Temp     0.6985414  0.2940876 -0.4971897  1.0000000
```

```r
# As seen before, and by previous scatter plots, Ozone is fairly correlated with
# the temperature and wind covariates.

### Simple Weighting
propensity_model <- glm(Observed~Wind+Temp,family=binomial(link="logit"), data=airquality)
response_probability <- predict(propensity_model, type='response')
response_weights <- 1 / response_probability
airquality$Weight <- response_weights
airquality$Weight[is.na(airquality$Ozone)] <- NA # set unobserved values to NA weight
sum(airquality$Weight,na.rm=T) # diagnostic is correct, sum(weights) = N
```

```
## [1] 152.9955
```

```r
airquality$Weight %>% hist(main="Histogram of Weights - Temp/Wind GLM")
```




```r
### Propensity Grouping (using fitted GLM)
airquality$propensity <- response_probability
prop_data <- airquality[order(airquality$propensity),c(1:4,9)] # sorted by propensity
propensity_groups <- split(prop_data,
                           cut(prop_data$propensity, breaks = 5, labels = FALSE))
# The values are so close together, groups aren't even. Even though they are not
# evenly sized, the proximity of the propensity values to one another immediately
# suggests this approach will not change much of the analysis. There are no
# cut-off points for the groups that correspond to a large jump in propensity,
```

```r
# so we expect each data point to have fairly similar weights, thus similar
# results to previous analysis

# this code is fairly confusing - it uses sapply on the split groups to
# calculate the weights based on # rows / # observed (i.e. not NA) for each
# group, then repeats each of those values according to the number of members in
# said group, setting the resulting vector of numbers as the Weight column
prop_data$Weight <- rep(sapply(propensity_groups,
                               function(data) data$Weight=nrow(data)/
                                 sum(!is.na(data$Ozone))),
                        sapply(propensity_groups,nrow))

rbind(Weighted_Mean=weighted.mean(airquality$Ozone, airquality$Weight, na.rm=T),
      Propensity_Groups=weighted.mean(prop_data$Ozone, prop_data$Weight, na.rm=T),
      Base_Mean=mean(airquality$Ozone,na.rm=T))
```

```
##                       [,1]
## Weighted_Mean     41.83034
## Propensity_Groups 42.14463
## Base_Mean         42.12931
```

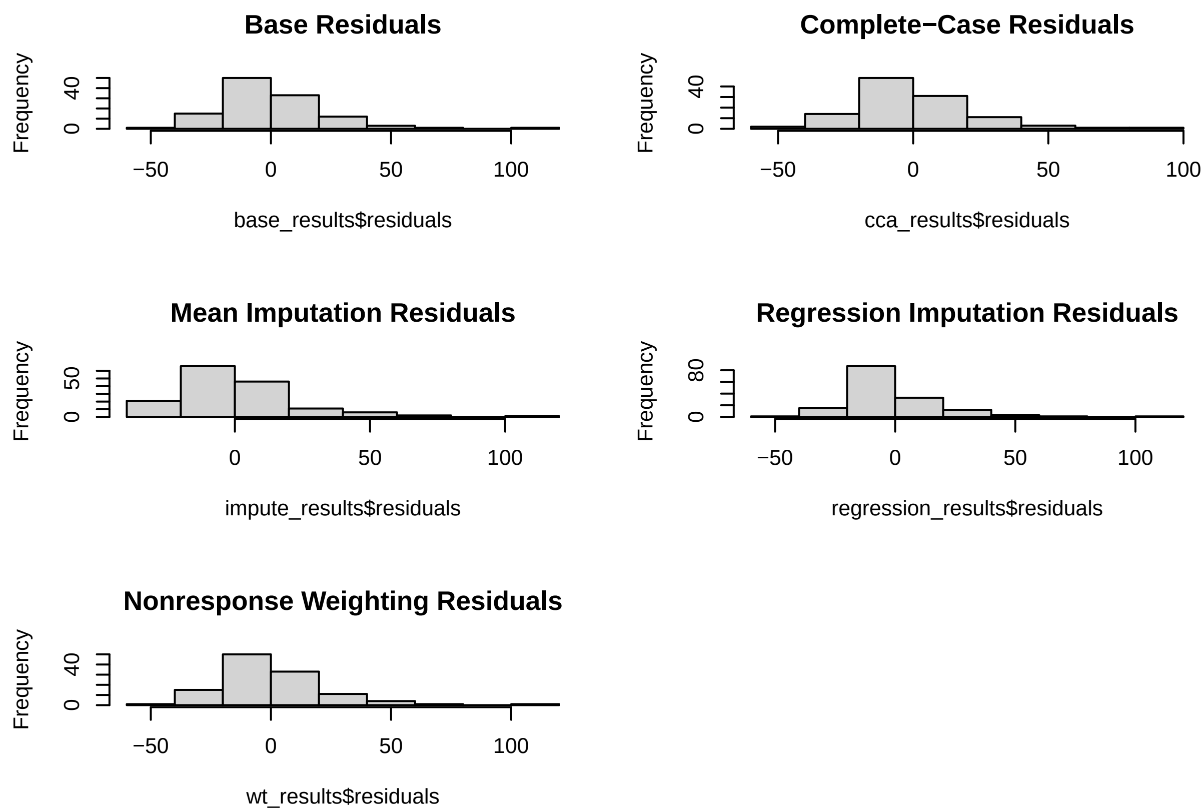
```r
# The values are extremely similar, because the weights for either method are
# extremely close together, so no particular value of Ozone is "favored" more
# than another.
# That said, insight into average ozone likely would not be of primary interest.
# Rather, a researcher is probably more interested in how the value changes over
# time, considering how the data set is a time series. Alternatively, they may
# be interested in the relationships among the variables in the data set, and
# thus a basic regression model is run below and compared to the regression
# results from ad-hoc methods.

complete_data <- airquality[!is.na(airquality$Ozone),]
wt_design <- svydesign(id=~1, weights=~Weight, data=complete_data)
wt_mean <- svymean(~Ozone, wt_design, na.rm=TRUE) # 41.8, same as above
wt_results <- svyglm(Ozone~Wind+Temp, wt_design)



# NOTE: All 5 methods had each parameter significant at 0.05 level, so the
# Std Errors, t values, and p-values are omitted here (but can be seen in the
# last problem)
rbind(Base=base_results$coefficients[,1],
      Complete_Case=cca_results$coefficients[,1],
      Mean_Imputation=impute_results$coefficients[,1],
      Regression_Imputation=regression_results$coefficients[,1],
      Nonresponse_Weighting=wt_results$coefficients)
```

```
##                       (Intercept) airquality$Wind airquality$Temp
## Base                    -71.03322       -3.055491        1.840179
## Complete_Case           -67.32195       -3.294839        1.827554
## Mean_Imputation         -41.21587       -2.598643        1.402387
## Regression_Imputation   -71.03322       -3.055491        1.840179
## Nonresponse_Weighting   -71.94309       -2.966150        1.840659
```

```r
# The biggest "outlier" in terms of model fitting is the mean imputation method.
# Each of its parameter estimates are closer to 0 than the alternative methods,
# which is to be expected from the attenuation discussed previously.
# On the other hand, the Base, Regression, and Nonresponse techniques were all
# extremely similar to one another - namely with the parameter estimate for
# temperature. Notably, the direction and magnitude of parameter estimates are
# very similar for all approaches, perhaps indicating that missingness is not
# a huge factor with this data set (at least for this particular analytical
# task). A key consideration is that the data set is technically a time series,
# and the passage of time likely plays some role in the values of either of the
# independent variables or the response.

par(mfrow=c(3,2))
hist(base_results$residuals,main="Base Residuals")
hist(cca_results$residuals,main="Complete-Case Residuals")
hist(impute_results$residuals,main="Mean Imputation Residuals")
hist(regression_results$residuals,main="Regression Imputation Residuals")
hist(wt_results$residuals,main="Nonresponse Weighting Residuals")
# The residuals all seem to follow similar shapes as well, though all notably
# have high-valued outliers on the right of their plots. The histograms appear
# to be relatively normal, and in any case statistical tests (K-S) conducted on
# any of the residual sets is likely to give very similar results to the others,
# because of how similar the distributions are.

# All of these results seem to suggest the data is very robust to any missing
# data techniques we have learned thus far. This could lead to the conclusion
# that missing values do not have a significant effect on the analysis, and
# we can likely conclude the Ozone variable is missing completely at random.
# This assertion is supported by the results of the t-tests in for Lecture 1
# Problem 3, which failed to detect any significant difference between the means
# of the explanatory variables between missing and observed responses. Though
# the correlation matrix and scatter plots revealed some relationship between
# Wind/Temp with Ozone, it could be that this relationship is not particularly
# strong, or explanatory "enough", for us to gain any additional insight with
# the missing data techniques used in this assignment. This is seen by the fact
# that the response weights all have fairly similar values - a variance near 0.
var(response_weights)
```

```
## [1] 0.001783489
```

```r
rm(airquality,base_results,cca_results,complete_data,impute_results, prop_data,
   propensity_groups,propensity_model,regression_results,wt_design,wt_results,
   response_probability,response_weights,wt_mean)
```



2

Read the gestational age dataset (dgestat.txt) and reproduce the statistics of Table 4 (ignoring some rounding error).

```
options(scipen=99,digits=3)
data <- read.table(file="C:/Users/jacma/OneDrive/school work/UMD/EPIB 664 Missing Data/dgestat.txt", co

definitions <- c("Gestational Age (weeks)","Birth Weight (grams)",
                 "Mother Age (years)","Father Age (years)",
                 "Apgar Score", "Mother Weight Gain (pounds)",
                 "# Prenatal Visits","# Cigarettes per day",
                 "# Drinks per week")
# using the apply function for efficiency
data.frame(Definitions = definitions,
           Mean=apply(data,2,mean,na.rm=T),
           Std.Dev=apply(data,2,sd,na.rm=T),
           Minimum=apply(data,2,min,na.rm=T),
           Maximum=apply(data,2,max,na.rm=T))
```

```
##                     Definitions      Mean Std.Dev Minimum Maximum
## dgestat     Gestational Age (weeks)   38.6678   2.081      19      44
## dbirwt         Birth Weight (grams) 3325.0639 585.838     227    5670
## dmage            Mother Age (years)   28.0681   5.976      13      49
```

```
## dfage            Father Age (years)   30.6012   6.762       15       67
## fmaps                   Apgar Score    8.9288   0.669        0       10
## wtgain  Mother Weight Gain (pounds)   31.0741  13.459        0       98
## nprevis           # Prenatal Visits   11.7598   3.669        0       49
## cigar          # Cigarettes per day    0.9975   3.749        0       60
## drink             # Drinks per week    0.0187   0.411        0       35
```

```
rm(data,definitions)
```